High-Dimensional Instrumental Variable Quantile Regression under Weak Instruments: A Monte Carlo Study

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Supervisor:
JProf. Claudia Noack Ph.D.

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Olimjon Umurzokov

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1 Introduction

Treatment effects need not be uniform across an outcome distribution. Education, training, or a policy intervention may affect outcomes differently for people near the lower, middle, and upper parts of that distribution. Quantile methods reveal this heterogeneity, which a mean effect can conceal (Koenker and Bassett 1978). A structural interpretation becomes difficult when treatment is endogenous. If education depends on unobserved ability or program participation on unobserved motivation, conventional quantile regression generally describes conditional associations rather than the effect of externally changing treatment. Instrumental variable quantile regression (IVQR) addresses this problem by combining instruments with structural quantile restrictions (Chernozhukov and Hansen 2005).

Modern applications may also contain many controls, transformations, and interactions. Their coefficients are nuisance parameters, but estimating them without structure can be unstable or infeasible. Regularization makes high-dimensional estimation manageable, while Neyman-orthogonal scores reduce the first-order sensitivity of the target moment to sufficiently small nuisance-estimation errors (Chernozhukov et al. 2018). Chen et al. (2021) combine these ideas with IVQR. Their DML-IVQR procedure accommodates high-dimensional controls and supports candidate-specific inference that does not require strong identification. Orthogonality, however, cannot create identifying information. A valid instrument may contain little information about the endogenous treatment, so moments can distinguish competing treatment-effect values only weakly.

This separation motivates the research gap. Existing DML-IVQR evidence shows how regularized nuisance estimation performs in a high-dimensional design with fixed instrument relevance, but it does not systematically trace how the same procedure behaves as excluded-instrument relevance declines while other structural features remain fixed. Weak-identification-robust inference can retain asymptotic validity without being informative: many false candidates may survive when the instruments provide little treatment variation. Its asymptotic reference distribution can also be imperfectly calibrated in moderate samples. The research question is therefore: How does weakening instrument strength affect the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how do these effects vary across quantiles?

The thesis answers this question through a controlled extension of the Monte Carlo design in Chen et al. (2021). A single parameter, κ, progressively reduces the loading of excluded-instrument primitives in the treatment equation. The extension preserves the marginal distribution of the latent treatment index, the endogeneity channel, the structural outcome equation, the quantile treatment effect, the controls, and the observed instrument construction. It therefore isolates designed changes in excluded-instrument relevance without treating κ as a universal measure of IVQR identification strength. The simulations compare an infeasible oracle benchmark, an unregularized full-control estimator, and DML-IVQR with regularized nuisance estimation on identical samples. Performance is assessed through point-estimation error, the size of numerically accepted sets, power against false candidate values, and coverage at the exact true parameter across several quantiles and two sample sizes.

The evidence gives a consistent answer. As relevance falls, point-estimation errors generally increase, accepted sets become less informative, and power declines. These responses differ across quantiles, with no universal ordering between central and tail quantiles. Coverage does not summarize informativeness and is not uniformly close to its nominal level in finite samples: candidate-specific inference may retain the truth while also retaining many false values, and the asymptotic approximation can under- or over-reject in particular design cells. Regularization often improves on estimating the full high-dimensional nuisance set without regularization, but DML does not dominate in every cell and cannot replace missing instrument information. The central implication is that nuisance robustness and identification robustness address distinct problems; neither guarantees precise or informative inference when excluded instruments contain little information about treatment.

Chapter 2 places this question in the literature, Chapters 3 and 4 present the method and simulation design, Chapters 5 and 6 report and interpret the evidence, and Chapter 7 concludes.

2 Econometric Background and Related Literature

Quantile regression describes how covariates are related to different parts of a conditional outcome distribution rather than only its mean (Koenker and Bassett 1978). This distinction matters when treatment effects are heterogeneous: an intervention may shift lower and upper quantiles by different amounts even when its average effect is modest. A structural quantile treatment effect compares the same quantile of potential-outcome distributions under alternative treatment states. It is generally a distributional effect, not an individual-level effect, because the observation occupying a given rank can differ across counterfactual states. Endogeneity creates a separate problem. When realized treatment depends on latent determinants of the outcome, the conditional distribution among observations selecting a treatment value need not equal the potential-outcome distribution obtained by fixing treatment at that value.

Ordinary quantile regression of the outcome on treatment and controls then estimates an observed conditional association and does not generally recover a structural quantile effect. The IVQR framework of Chernozhukov and Hansen (2005) uses excluded instruments to connect endogenous treatment variation to structural outcome quantiles.

Structural IVQR represents potential outcomes through a quantile function and latent ranks. Instrument exogeneity restricts the relationship between the instruments and these structural ranks conditional on controls, while treatment may still depend on latent variables related to the outcome. Rank invariance assumes that an individual's rank is unchanged across treatment states; the weaker rank-similarity condition allows ranks to change but requires their conditional distributions to agree across those states. Under the maintained selection, exogeneity, and rank conditions, the probability that the observed outcome lies below the true structural quantile, conditional on controls and instruments, equals the quantile index (Chernozhukov and Hansen 2005; Chernozhukov et al. 2017). This conditional restriction links the potential-outcome model to estimable instrument moments. Validity permits the instrument to enter those moments; relevance determines whether instrument-induced treatment variation separates competing structural coefficients. Moment validity alone does not establish point identification. A unique solution additionally requires the population mapping from candidate structural quantiles to the restrictions to distinguish the truth from alternatives. That mapping is quantile specific and density weighted, so a mean first-stage statistic is informative about relevance but does not completely characterize IVQR identification at a particular quantile. This distinction is essential empirically: an instrument can satisfy the maintained exogeneity condition yet generate too little treatment variation to locate a structural quantile effect precisely.

When identifying information is strong, the population moment moves sufficiently as the candidate treatment coefficient moves away from its true value, supporting regular point-estimator approximations. With weak information, the true value may remain unique while nearby candidates generate almost indistinguishable moments. Partial identification instead permits more than one population solution, and failed identification leaves some candidates or directions indistinguishable. Chernozhukov and Hansen (2008) develop inference that remains asymptotically valid without imposing the strong-identification conditions needed for conventional estimate-centered inference. Their approach fixes a candidate value, estimates or profiles the nuisance parameters under that candidate, and tests whether the resulting instrument-based quantile moments are compatible with zero. Inverting these candidate-specific tests retains every non-rejected value and can therefore produce asymmetric, disconnected, or very wide confidence regions. Weak-identification robustness concerns the null reference distribution; it does not guarantee that false candidates will be rejected. Informativeness and finite-sample calibration remain distinct: a robust confidence region can honestly reveal little about the parameter, and an asymptotic test can still reject too often or too rarely in a finite sample. The exact conditional procedure of Chernozhukov et al. (2009) uses a different simulated reference distribution and is not the method implemented here.

High-dimensional controls create a second challenge because estimating the outcome and instrument nuisances may require many coefficients relative to the sample size. Regularization exploits approximate sparsity: a comparatively small set of variables captures most predictive content, while the remaining coefficients are zero or sufficiently small. Shrinkage and selection can nevertheless contaminate a target moment. Double machine learning addresses this problem with a Neyman-orthogonal score whose population expectation has zero first derivative with respect to local nuisance perturbations at the truth (Chernozhukov et al. 2018). Orthogonality therefore removes the leading sensitivity to sufficiently small nuisance-estimation errors, not the errors themselves; the learners must still satisfy suitable convergence and regularity conditions. Sample splitting estimates nuisances on one subsample and evaluates the score on another, while cross-fitting rotates these roles across folds. These concepts should not be conflated: orthogonality is a property of the score, whereas cross-fitting organizes data use and limits overfitting dependence. Nor does DML strengthen an instrument. It addresses high-dimensional nuisance estimation, while candidate-specific robust inference addresses irregular inference under limited identification.

Chen et al. (2021) join these two strands. They profile a candidate-specific high-dimensional quantile regression, construct a density-weighted instrument residual, form a Neyman-orthogonal IVQR score, estimate the nuisances with regularization, and evaluate candidates by GMM and weak-identification-robust test inversion. Although they discuss cross-fitting as part of the broader DML framework, the executable DML-IVQR algorithm preserved in this thesis estimates nuisances and evaluates the score on the same sample. Their principal Monte Carlo design holds instrument relevance fixed while comparing Oracle-GMM, Full-GMM, and DML-IVQR across several quantiles. The remaining empirical question is how point-estimation error, confidence-set informativeness, power, and finite-sample calibration respond when excluded-instrument relevance is systematically weakened within that otherwise preserved design. This thesis supplies that controlled stress test; it does not propose a new structural model, score, estimator, identification theorem, or finite-sample correction. Jin and Sun (2025) study a related nonparametric high-dimensional IVQR problem for unconditional quantile treatment effects with automatic DML and cross-fitting, but their target and design differ from the controlled relevance experiment considered here. Chapter 3 states the exact structural model, nuisance construction, estimators, and inference procedure used in the simulations.

3 DML-IVQR Methodology

3.1 Structural IVQR Model and Identification

Following the instrumental-variable quantile regression (IVQR) framework of Chernozhukov and Hansen (2005) and Chernozhukov and Hansen (2008), let Y_i denote a scalar outcome, D_i an endogenous treatment or vector of endogenous variables, X_i a vector of observed controls, and Z_i a vector of instruments. For each potential treatment state d, the potential outcome satisfies

$$Y_{di} = q(U_{di}, d, X_i), \qquad U_{di} \mid X_i \sim \text{Uniform}(0, 1), \tag{1}$$

where U_{di} is the treatment-state-specific structural rank and $q(u, d, x)$ is strictly increasing in u. Thus, $q(\tau, d, x)$ is the conditional τ-quantile of the potential outcome Y_d given $X = x$, and hence the structural quantile obtained by fixing treatment at d. The IVQR assumptions combine conditional instrument exogeneity for the potential-outcome ranks, a treatment-selection mechanism, and rank similarity across treatment states. Define the realized structural rank as $U_i := U_{D_i i}$. Rank similarity does not impose the stronger special case of rank invariance, under which $U_{di} = U_{d'i}$ for all treatment states d and d'. A central implication of these assumptions is the observed-data representation

$$Y_i = q(U_i, D_i, X_i), \qquad U_i \mid X_i, Z_i \sim \text{Uniform}(0, 1). \tag{2}$$

Here U_{di} is the rank under potential treatment d, whereas U_i is the rank realized under D_i; rank similarity does not require identical individual ranks across treatment states.

Endogeneity permits dependence between D_i and U_i. The observed conditional quantile $Q_{Y|D,X}(\tau \mid d, x)$ then reflects selection into treatment and generally differs from the structural quantile $q(\tau, d, x)$ obtained by fixing treatment. For states d_1 and d_0, their structural quantile treatment effect at x is $q(\tau, d_1, x) - q(\tau, d_0, x)$.

For a fixed $\tau \in (0, 1)$, the DML-IVQR procedure of Chen et al. (2021) adopts the linear-in-parameters structural quantile specification

$$q(\tau, D_i, X_i) = c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau). \tag{3}$$

Here $c_0(\tau)$ is the intercept, $\alpha_0(\tau)$ is the low-dimensional structural parameter of interest, and the potentially high-dimensional $\beta_0(\tau)$ contains nuisance control slopes. With scalar D_i, $\alpha_0(\tau)$ is the structural quantile effect per unit of treatment.

Strict monotonicity in the rank gives $\{Y_i \leq q(\tau, D_i, X_i)\} = \{U_i \leq \tau\}$. Together with the conditional uniformity in (2), this yields the key restriction at the true structural parameters,

$$\Pr\left[Y_i \leq c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau) \mid X_i, Z_i\right] = \tau. \tag{4}$$

Equivalently, the model implies the conditional moment

$$\mathrm{E}\left[\tau - \mathbb{1}\left\{Y_i \leq c_0(\tau) + D_i'\alpha_0(\tau) + X_i'\beta_0(\tau)\right\} \,\middle|\, X_i, Z_i\right] = 0. \tag{5}$$

Thus the true structural residual is nonpositive with probability τ conditional on (X_i, Z_i). This population restriction identifies $\alpha_0(\tau)$ when it uniquely distinguishes the truth from alternatives; relevance governs how sharply it does so. The next subsection profiles the high-dimensional nuisance for each candidate.

3.2 Orthogonal Score and Nuisance Estimation

Fix $\tau \in (0, 1)$ and let a denote a candidate value for the low-dimensional structural coefficient $\alpha_0(\tau)$. Define the augmented control vector $\widetilde{X}_i = (1, X_i')'$, whose first element represents the intercept and whose remaining elements are the nonconstant controls. Profiling temporarily treats a as fixed. After subtracting $D_i'a$ from Y_i, the intercept and control slopes are chosen together to fit the τ-quantile of the remaining outcome according to the quantile check loss

$$\rho_\tau(u) = u\left[\tau - \mathbb{1}\{u < 0\}\right]. \tag{6}$$

The check loss gives τ-specific asymmetric weights to positive and negative residuals. Following Chen et al. (2021), define

$$\begin{pmatrix} c(a) \\ \beta(a) \end{pmatrix} = \arg\min_{c,b} \mathrm{E}\left[\rho_\tau\left(Y_i - D_i'a - c - X_i'b\right)\right]. \tag{7}$$

These population coefficients change with the adjusted outcome $Y_i - D_i'a$ and satisfy

$$\mathrm{E}\left[\left\{\tau - \mathbb{1}\left[Y_i \leq D_i'a + c(a) + X_i'\beta(a)\right]\right\} \widetilde{X}_i\right] = 0. \tag{8}$$

The constant and remaining elements of $\widetilde{X}_i$ give the intercept and slope conditions. At $a = \alpha_0(\tau)$, maintained uniqueness gives $c(a) = c_0(\tau)$ and $\beta(a) = \beta_0(\tau)$; otherwise they are candidate-specific profile nuisances.

The associated profile residual is

$$\epsilon_i(a) = Y_i - D_i'a - c(a) - X_i'\beta(a). \tag{9}$$

Let $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ be its conditional density at zero. This density governs how many observations cross the candidate quantile after a small change in $c(a)$ or $\beta(a)$.

To characterize this sensitivity, suppose $Z_i \in \mathbb{R}^{d_z}$ and $X_i \in \mathbb{R}^p$, so $\widetilde{X}_i \in \mathbb{R}^{p+1}$, and define

$$\begin{aligned} M(a) &= \mathrm{E}\big[Z_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\big], \quad M(a) \in \mathbb{R}^{d_z \times (p+1)}, \\ J(a) &= \mathrm{E}\big[\widetilde{X}_i\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)\big], \quad J(a) \in \mathbb{R}^{(p+1)\times(p+1)}. \end{aligned} \tag{10}$$

$J(a)$ records density-weighted augmented-control variation, while $M(a)$ records the corresponding instrument–control relationship. Assuming $J(a)$ is nonsingular, set

$$\delta(a) = M(a)J(a)^{-1}, \qquad \delta(a) \in \mathbb{R}^{d_z \times (p+1)}, \tag{11}$$

Partition this matrix as $\delta(a) = [d_0(a), \Gamma(a)]$, where $d_0(a) \in \mathbb{R}^{d_z}$ contains the residualization intercepts and $\Gamma(a) \in \mathbb{R}^{d_z \times p}$ contains the slopes on the nonconstant controls. The corresponding population residualized instrument is

$$\Psi_i(a) = Z_i - \delta(a)\widetilde{X}_i = Z_i - d_0(a) - \Gamma(a)X_i. \tag{12}$$

Thus $\Psi_i(a)$ removes the density-weighted instrument component associated with the intercept and controls. By construction, $\mathrm{E}[(Z_i - \delta(a)\widetilde{X}_i)\widetilde{X}_i' f_{\epsilon(a)}(0 \mid X_i, Z_i)] = 0$; this is not an ordinary unweighted residual or a conventional first-stage coefficient.

Collect the candidate-specific nuisance quantities as

$$\eta(a) = \begin{pmatrix} c(a) \\ \beta(a) \\ \operatorname{vec}\{\delta(a)\} \end{pmatrix}, \qquad \eta_0 = \eta(\alpha_0(\tau)) = \begin{pmatrix} c_0(\tau) \\ \beta_0(\tau) \\ \operatorname{vec}\{\delta(\alpha_0(\tau))\} \end{pmatrix}. \tag{13}$$

Vectorization only stacks the residualization matrix. The target remains $\alpha_0(\tau)$; $c(a)$ and $\beta(a)$ profile the outcome nuisance, while $\delta(a)$ residualizes the instruments.

Following Chen et al. (2021), combine the nuisance quantities above in the score

$$\begin{aligned} g_\tau(W_i; a, \eta) &= \big[\tau - \mathbb{1}\{Y_i \le D_i'a + c + X_i'\beta\}\big]\left(Z_i - \delta\widetilde{X}_i\right), \\ g_\tau(W_i; a, \eta(a)) &= \big[\tau - \mathbb{1}\{Y_i \le D_i'a + c(a) + X_i'\beta(a)\}\big]\,\Psi_i(a), \end{aligned} \tag{14}$$

where $W_i = (Y_i, D_i, X_i, Z_i)$. The first factor is the quantile score: it equals τ above the fitted quantile and $\tau - 1$ at or below it. The second is the density-weighted residualized instrument. Multiplying them gives the IVQR moment used to assess a.

At $a = \alpha_0(\tau)$ and $\eta = \eta_0$, the score satisfies

$$\mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta_0)] = 0. \tag{15}$$

The conditional moment in Section 3.1 remains zero after multiplication by a function of (X_i, Z_i), giving (15). Moment validity follows from the structural assumptions; Neyman orthogonality is the separate nuisance property

$$\partial_\eta \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} = 0. \tag{16}$$

Small nuisance errors therefore have no first-order effect on the population moment at the truth. Orthogonality reduces the effect of nuisance-estimation error; it does not remove that error.

To show the cancellation for the intercept and slopes, write $\epsilon_{0i} = \epsilon_i(\alpha_0(\tau))$ and define

$$\begin{aligned} M_0 &= \mathrm{E}\big[Z_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ J_0 &= \mathrm{E}\big[\widetilde{X}_i \widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big], \\ \delta_0 &= M_0 J_0^{-1}. \end{aligned} \tag{17}$$

At the truth, differentiation with respect to the fitted intercept and control slopes then gives

$$\begin{aligned} \partial_{(c,\beta')} \mathrm{E}[g_\tau(W_i; \alpha_0(\tau), \eta)]\big|_{\eta=\eta_0} &= -\mathrm{E}\big[(Z_i - \delta_0 \widetilde{X}_i)\widetilde{X}_i' f_{\epsilon_0}(0 \mid X_i, Z_i)\big] \\ &= -[M_0 - \delta_0 J_0] = 0. \end{aligned} \tag{18}$$

The density appears because observations near the boundary can switch sides. The choice $\delta_0 = M_0 J_0^{-1}$ makes the residualized instrument density-weighted orthogonal to $\widetilde{X}_i$, cancelling first-order changes in the intercept and slopes.

Orthogonality also holds in the δ direction. For instrument component j, let δ_j denote the corresponding row of δ. Then

$$\partial_{\delta_j} \mathrm{E}\big[g_{\tau,j}(W_i; \alpha_0(\tau), \eta)\big]\big|_{\eta=\eta_0} = -\mathrm{E}\big[\big\{\tau - \mathbb{1}\big[Y_i \le D_i'\alpha_0(\tau) + c_0(\tau) + X_i'\beta_0(\tau)\big]\big\}\widetilde{X}_i'\big] = 0. \tag{19}$$

An error in δ is multiplied by the quantile residual, whose moment with $\widetilde{X}_i$ is zero by (8), so its first-order effect also vanishes. This construction permits sufficiently accurate regularized nuisances to support inference on the low-dimensional target (Chernozhukov et al. 2018; Chen et al. 2021).

Orthogonality protects the score against small nuisance-parameter errors, but it does not create identification or add information about the treatment effect. In particular, it does not make a weak instrument strong and does not guarantee that the data contain enough information about $\alpha_0(\tau)$. If the instruments contain little information about the endogenous treatment, the moment may also contain little information for distinguishing nearby values of a. IVQR moment validity comes from the structural assumptions and instrument exogeneity; nuisance robustness comes from orthogonalization. These are different requirements.

3.3 Estimators and Executable Implementation

Section 3.2 defined population nuisances. When controls are numerous relative to N, DML-IVQR estimates their sample counterparts with ℓ_1 regularization under approximate sparsity: a relatively small group has substantial predictive content while other coefficients are zero or small.

Fix τ and a candidate value a. Nuisance estimation is repeated whenever a relevant candidate is evaluated. In the executable implementation used here, the response for the first nuisance step is $Y_i - D_i'a$. The formula adds an intercept, so let b_0 denote that intercept and let $b = (b_1, \ldots, b_p)'$ contain only the slopes on X_i. Thus, $(\widehat{b}_0(a), \widehat{\beta}(a))$ estimates $(c(a), \beta(a))$. The implementation-specific notation for the later instrument regression is defined below and is kept distinct from the full population matrix $\delta(a) = [d_0(a), \Gamma(a)]$. The candidate-specific quantile-Lasso estimates are

$$\begin{pmatrix} \widehat{b}_0(a) \\ \widehat{\beta}(a) \end{pmatrix} = \arg\min_{b_0, b} \left\{ \sum_{i=1}^{N} \rho_\tau(Y_i - D_i'a - b_0 - X_i'b) + \lambda_0(a)|b_0| + \sum_{k=1}^{p} \lambda_k(a)|b_k| \right\}. \tag{20}$$

The first term measures fit at quantile τ. The second places a separate penalty on each coefficient, including the intercept; there is no post-Lasso refit. The loss is an unscaled sum because this is the scaling used by the fitted quantile-regression routine. Following the pivotal route described by Chen et al. (2021), let $s_k = (N^{-1} \sum_i X_{ik}^2)^{1/2}$ and, for $r = 1, \ldots, 1000$, draw independent $U_{ir} \sim \text{Uniform}(0, 1)$. The implemented penalty is

$$\begin{aligned} T_r &= \max_{1 \le k \le p} \left| \frac{\sum_{i=1}^{N} X_{ik} \left[\tau - \mathbb{1}\{U_{ir} < \tau\}\right]}{s_k \sqrt{\tau(1-\tau)}} \right|, \\ \lambda(a) &= 2\, \widehat{Q}_{0.9}\left(\{T_r\}_{r=1}^{1000}\right) \sqrt{\tau(1-\tau)}(1, s_1, \ldots, s_p)'. \end{aligned} \tag{21}$$

Artificial Uniform variables are useful because the indicator at the correct quantile has a known Bernoulli structure. The simulated statistics approximate how large random correlations between the controls and a quantile score can be without a true signal. The simulated 0.9 quantile sets the reference level for these random correlations, while the factor 2 scales the resulting penalty according to the implemented pivotal rule. A fresh set of Uniform draws is generated every time the penalty routine is called. Hence, candidate evaluations on the same observed sample can use different pivotal draws. This is penalty-construction randomness, not sample splitting. The frozen main simulation uses this rule rather than its older five-fold cross-validation alternative.

The fitted quantile regression produces the candidate residual and density weight

$$\begin{aligned} \widehat{\epsilon}_i(a) &= Y_i - D_i'a - \widehat{b}_0(a) - X_i'\widehat{\beta}(a), \\ \widehat{f}_i(a) &= \phi\Big(\widehat{\epsilon}_i(a); \overline{\widehat{\epsilon}}(a), s^2_{\widehat{\epsilon}(a)}\Big), \end{aligned} \tag{22}$$

where $\phi(e;\mu,\sigma)$ is the normal density with mean μ and standard deviation σ. Thus, the executable simulation forms the weights by evaluating a fitted normal density at the candidate residuals.[1] This fitted-normal quantity is an implementation-specific proxy for the theoretical conditional density $f_{\epsilon(a)}(0 \mid X_i, Z_i)$ from Section 3.2; it is not a direct conditional-density estimate at zero. Zero remains the fitted quantile boundary, and the theoretical reason for density weighting is that observations near this boundary determine local changes in the quantile moment. These proxy weights are then used in the instrument-residualization step, where they play the role assigned to density weights in the theoretical construction.

For each instrument component j, the implementation next fits a separate square-root-density-weighted Lasso. Let $\widehat{r}_{j0}(a)$ denote its retained ordinary intercept and let $\widehat{\Gamma}_{\mathrm{imp},j}(a)$ contain only the slopes on the transformed controls. Its transformed regression is

$$\begin{pmatrix} \widehat{r}_{j0}(a) \\ \widehat{\Gamma}_{\mathrm{imp},j}(a) \end{pmatrix} = \underset{(r_{j0},\gamma_j)}{\arg\min} \left\{ \sum_{i=1}^{N} \left[\sqrt{\widehat{f}_i(a)} Z_{ji} - r_{j0} - \sqrt{\widehat{f}_i(a)} X_i'\gamma_j \right]^2 + \mathcal{P}_{j,a}(\gamma_j) \right\}. \tag{23}$$

Here $\mathcal{P}_{j,a}(\gamma_j) = \sum_{k=1}^{p} \lambda^{\Gamma}_{jk}(a)|\gamma_{jk}|$ is the coefficient-specific ℓ_1 penalty produced by the instrument-regression routine's default iterative heteroskedastic loadings. The intercept is included but not penalized, the same controls are used as in the first nuisance step, and no post-Lasso refit is performed. After fitting the transformed regression, the code retains both the

1. The literal preserved call is `dnorm(e, mean(e), var(e))`. Its third argument is interpreted as a standard deviation, although the code supplies the residual sample variance.

intercept and the slopes. Define $\widehat{r}_0(a) = (\widehat{r}_{10}(a), \ldots, \widehat{r}_{d_z 0}(a))'$ and stack the slope rows in $\widehat{\Gamma}_{\text{imp}}(a)$. The residualized instrument used in the score is

$$\begin{aligned} \widehat{\Psi}_{ji}(a) &= Z_{ji} - \widehat{r}_{j0}(a) - X_i' \widehat{\Gamma}_{\text{imp},j}(a), \\ \widehat{\Psi}_i(a) &= Z_i - \widehat{r}_0(a) - \widehat{\Gamma}_{\text{imp}}(a) X_i. \end{aligned} \tag{24}$$

Thus, the score residualizes the original, untransformed instrument. The calculation subtracts the retained intercept and the fitted contribution of the original controls. It does not divide the retained intercept by $\sqrt{\widehat{f}_i(a)}$ or use the transformed regression residual directly.

The intercept retained by the frozen transformed regression is implementation specific. It enters as an ordinary constant rather than $\sqrt{\widehat{f}_i(a)}$ times a constant regressor, so $\widehat{r}_0(a)$ is not the literal sample counterpart of $d_0(a)$ in the augmented population projection of Section 3.2. The exact derivative cancellation in Section 3.2 applies to the theoretical augmented score. The executable score follows its density-weighted slope residualization but treats the intercept differently. The results therefore characterize the preserved implementation, not a literal sample implementation of every component of the theoretical score.

The penalties can shrink weak coefficients to zero and stabilize both high-dimensional steps, but also create shrinkage error. The theoretical score in Section 3.2 limits its first-order effect (Chernozhukov et al. 2018). Nuisances and scores use the same sample; this preserved implementation has no sample splitting or cross-fitting.

Oracle-GMM, Full-GMM, and DML-IVQR target the same $\alpha_0(\tau)$ using the same outcome, treatment, instruments, and quantile. Each multiplies a candidate quantile score by residualized instruments, producing a moment with dimension d_z; nuisance estimation differs.

Oracle-GMM is an infeasible benchmark because it is given the identity of the ten controls designated as relevant in the simulation DGP. Let $S_0 = \{1, \ldots, 10\}$ denote this oracle control set and let $X_{S_0,i}$ collect these variables. At each candidate, the executable estimator uses an unpenalized quantile regression with an intercept:

$$\begin{pmatrix} \widehat{b}_{0,\text{orc}}(a) \\ \widehat{\beta}_{\text{orc}}(a) \end{pmatrix} = \operatorname*{arg\,min}_{b_0, b} \sum_{i=1}^{N} \rho_\tau \Big(Y_i - D_i' a - b_0 - X_{S_0,i}' b \Big) . \tag{25}$$

Thus, Oracle-GMM first removes $D_i' a$ and then fits the remaining outcome at quantile τ using only this known oracle control set. No coefficient is penalized and there is no later variable-selection or refitting step. The oracle is not a practically available estimator and is not free from sampling error. It provides a benchmark for the cost of having to learn which controls matter.

Full-GMM does not receive this knowledge. It uses all available controls in the same unpenalized, candidate-specific quantile-regression problem:

$$\begin{pmatrix} \widehat{b}_{0,\mathrm{full}}(a) \\ \widehat{\beta}_{\mathrm{full}}(a) \end{pmatrix} = \arg\min_{b_0,b} \sum_{i=1}^{N} \rho_\tau\left(Y_i - D_i'a - b_0 - X_i'b\right). \tag{26}$$

Thus all 100 controls enter freely. Full-GMM does not require the unknown oracle set, but estimating many weak or irrelevant coefficients can be noisy.

Oracle-GMM and Full-GMM use the same unregularized instrument residualization with their respective control matrices. For $e \in \{\mathrm{orc}, \mathrm{full}\}$, let $X_{\mathrm{orc},i} = X_{S_0,i}$ and $X_{\mathrm{full},i} = X_i$, and let $\widehat{f}_{e,i}(a)$ be the fitted-normal proxy defined above. The code computes

$$\begin{aligned} \widehat{M}_e(a) &= \frac{1}{N}\sum_{i=1}^{N} Z_i X_{e,i}' \widehat{f}_{e,i}(a), \qquad & \widehat{J}_e(a) &= \frac{1}{N}\sum_{i=1}^{N} X_{e,i} X_{e,i}' \widehat{f}_{e,i}(a), \\ \widehat{\delta}_e(a) &= \widehat{M}_e(a)\widehat{J}_e(a)^{-1}, & \widehat{\Psi}_{e,i}(a) &= Z_i - \widehat{\delta}_e(a) X_{e,i}. \end{aligned} \tag{27}$$

This direct, unpenalized inversion includes no residualization intercept, although the quantile regression has one, and uses the original Z_i in the final residual. It therefore differs from the augmented population orthogonalization in Section 3.2. There is no ridge adjustment or generalized inverse: a singular $\widehat{J}_e(a)$ produces a recorded numerical failure. Each benchmark multiplies $\widehat{\Psi}_{e,i}(a)$ by its candidate quantile score.

DML-IVQR uses all controls and regularizes both nuisance steps. It estimates the control coefficients by quantile Lasso, forms the density proxy, and uses regularized weighted instrument regressions. DML-IVQR-BC applies (21). Oracle and Full also use residualized IVQR moments, but estimate nuisances without ℓ_1 regularization. Unlike their matrix calculation, DML retains a residualization intercept. The comparison covers a known oracle set without regularization, all controls without regularization, and all controls with regularized nuisance estimation. It does not isolate every implementation difference but keeps the structural model, target, and instruments common.

3.4 GMM Criterion and Weak-Identification-Robust Inference

Let $e \in \{\mathrm{orc}, \mathrm{full}, \mathrm{dml}\}$ index the estimator and $\widehat{g}_{e,i}(a) \in \mathbb{R}^{d_z}$ its observation-level score. Its sample average is

$$\widehat{g}_{e,N}(a) = \frac{1}{N}\sum_{i=1}^{N} \widehat{g}_{e,i}(a). \tag{28}$$

This vector measures the sample restriction at a. The code measures its variation with the uncentered second moment

$$\widehat{\Sigma}_e(a) = \frac{1}{N}\sum_{i=1}^{N}\widehat{g}_{e,i}(a)\widehat{g}_{e,i}(a)'. \tag{29}$$

This matrix standardizes differently scaled and correlated components. It uses denominator N, no finite-sample correction, and no sample-mean centering. At the truth, its probability limit equals the score covariance because the population moment is zero.

The common candidate-specific GMM criterion is

$$W_{e,N}(a) = N\widehat{g}_{e,N}(a)'\widehat{\Sigma}_e(a)^{-1}\widehat{g}_{e,N}(a). \tag{30}$$

This equals the code's calculation from the unnormalized score sum and sum of outer products. Smaller values indicate moments closer to zero after standardization. The direct inverse has no ridge or generalized inverse; failure or a nonfinite result is recorded. Suppressing e gives $W_N(a)$. The point estimator is

$$\widehat{\alpha}_e(\tau) = \arg\min_{a\in\mathcal{A}} W_{e,N}(a). \tag{31}$$

The minimum need not be zero. The implementation minimizes over the finite profile in Section 4.3, selects the first exact tie, and gives no point estimate to an incomplete profile.

Point estimation retains the minimizer; inference tests each a separately. Under the regularity conditions for the corresponding theoretical candidate-specific IVQR score, the null statistic has the asymptotic reference distribution

$$W_{e,N}(a) \xrightarrow{d} \chi^2_{d_z}, \qquad d_z = \dim(Z_i). \tag{32}$$

The degrees of freedom equal the number of moment components. The preserved Monte Carlo implementation applies this chi-square law as the asymptotic reference distribution for the candidate-specific statistics of all three estimators. For DML-IVQR-BC, however, the implementation-specific residualization intercept described in Section 3.3 is not the literal sample counterpart of the augmented theoretical intercept. This thesis therefore does not establish a separate limit theorem for that implementation detail. Following Chernozhukov and Hansen (2008), the test evaluates the hypothesized value directly rather than using a normal approximation for $\widehat{\alpha}_e(\tau)$. At level γ, reject when $W_{e,N}(a) > c_{1-\gamma,d_z}$.

Inverting these candidate-specific tests gives

$$C_{e,1-\gamma}(\tau) = \left\{a \in \mathcal{A} : W_{e,N}(a) \le c_{1-\gamma,d_z}\right\}. \tag{33}$$

Test inversion retains every candidate not rejected by its own moment test. Because the profile can cross the cutoff repeatedly, the region may be connected, disconnected, wide, or empty. A finite numerical domain describes only acceptance within that domain and cannot establish global unboundedness.

The underlying Chernozhukov–Hansen test-inversion approach can retain asymptotic validity under weak, partial, or failed identification, subject to its regularity conditions, but need not be informative. Weak instruments can leave many candidates below the cutoff. Because the implementation uses an asymptotic chi-square critical value, finite-sample calibration remains empirical. This differs from the exact conditional simulation method of Chernozhukov et al. (2009).

4 Monte Carlo Design

Chapter 3 uses N as generic sample-size notation; the simulation chapters use n for the design sample size.

4.1 Data-Generating Process and Instrument-Relevance Extension

The Monte Carlo experiment builds on the high-dimensional simulation design of Chen et al. (2021), with $p = 100$ observed controls. For each observation i, the structural disturbances satisfy

$$\begin{pmatrix} u_i \\ \epsilon_i \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}\right). \tag{34}$$

The disturbance ϵ_i enters the treatment equation, whereas u_i enters the outcome through $D_i u_i$. Their positive correlation therefore generates endogeneity.

The remaining primitives are mutually independent and independent of the disturbance pair in Equation (34), and they satisfy

$$x_{ji} \overset{\text{ind}}{\sim} \mathcal{N}(0,1), \quad j = 1, \dots, 100, \qquad z_{1i}, z_{2i}, v_{1i}, v_{2i} \overset{\text{ind}}{\sim} \mathcal{N}(0,1). \tag{35}$$

Observations are independent across i. The observed controls are

$$X_{ji} = \Phi(x_{ji}), \qquad j = 1, \dots, 100, \tag{36}$$

where Φ is the standard-normal cumulative distribution function, so $X_{ji} \sim \mathcal{U}(0,1)$ independently across controls.

The executable design constructs two instruments as[2]

$$Z_{1i} = z_{1i} + v_{1i} + X_{2i} + X_{3i} + X_{4i}, \tag{37}$$

$$Z_{2i} = z_{2i} + v_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i}. \tag{38}$$

The z components enter the treatment index directly, the included controls induce dependence between instruments and controls, and the v components add independent noise. The executable baseline treatment is

$$D_i = \Phi(z_{1i} + z_{2i} + \epsilon_i). \tag{39}$$

The treatment is continuous and lies in $(0,1)$.

The structural outcome is generated according to

$$Y_i = 1 + D_i + 5\sum_{j=1}^{7} X_{ji} + D_i u_i. \tag{40}$$

The outcome contains $X_{1i}, \ldots, X_{7i}$; together with the controls in the instruments, $X_{1i}, \ldots, X_{10i}$ form the ten-control oracle set. For $d \in (0,1)$, Equation (40) implies

$$Y_i(d) = 1 + 5\sum_{j=1}^{7} X_{ji} + d(1 + u_i). \tag{41}$$

Because u_i is independent of X_i and standard normal,

$$Q_{Y_i(d)|X_i}(\tau) = 1 + 5\sum_{j=1}^{7} X_{ji} + d\left[1 + \Phi^{-1}(\tau)\right]. \tag{42}$$

The structural quantile treatment effect is therefore

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau). \tag{43}$$

In particular, $\alpha_0(0.50) = 1$.

2. The displayed Monte Carlo equations in Chen et al. (2021) write the corresponding control terms as x_j. The released simulation code instead uses the transformed controls $X_j = \Phi(x_j)$. The simulations in this thesis follow that executable implementation.

The extension varies excluded-instrument relevance through $\kappa \in [0, 1]$. Let $w_i \sim \mathcal{N}(0, 1)$ be independent of all preceding primitives. The latent index and treatment are

$$d_i^*(\kappa) = \kappa(z_{1i} + z_{2i}) + \sqrt{2(1 - \kappa^2)}\, w_i + \epsilon_i, \tag{44}$$

$$D_i(\kappa) = \Phi\big(d_i^*(\kappa)\big)\,. \tag{45}$$

This symmetric loading is the sole structural modification to the executable baseline.

At $\kappa = 1$, the coefficient on w_i is zero and

$$D_i(1) = \Phi(z_{1i} + z_{2i} + \epsilon_i), \tag{46}$$

which reproduces Equation (39). Independence and unit variances imply

$$\mathrm{Var}\big[d_i^*(\kappa)\big] = 2\kappa^2 + 2(1 - \kappa^2) + 1 = 3. \tag{47}$$

Thus $d_i^*(\kappa) \sim \mathcal{N}(0, 3)$ for every κ, preserving the marginal distribution of $D_i(\kappa)$. This distribution is not uniform because Φ is applied to an index with variance three.

Instrument relevance changes because

$$\mathrm{Cov}\big(Z_{1i}, d_i^*(\kappa)\big) = \mathrm{Cov}\big(Z_{2i}, d_i^*(\kappa)\big) = \kappa. \tag{48}$$

All other instrument components are independent of $d_i^*(\kappa)$, so lower κ reduces each instrument–index covariance. The parameter governs relevance but is not a complete measure of IVQR identification. Endogeneity remains fixed because u_i is independent of z_{1i}, z_{2i}, and w_i while its covariance with ϵ_i is unchanged:

$$\mathrm{Cov}\big(u_i, d_i^*(\kappa)\big) = \mathrm{Cov}(u_i, \epsilon_i) = 0.3. \tag{49}$$

Joint normality and the invariant variance and covariance in Equations (47) and (49) keep the joint distribution of $(u_i, d_i^*(\kappa))$ unchanged. The controls, observed-instrument construction, structural outcome equation, induced marginal outcome distribution, and structural quantile effect are also unchanged. The extension therefore isolates relevance from the marginal treatment distribution, endogeneity channel, and structural treatment-effect distribution.

4.2 Simulation Design and Estimator Comparison

The completed Monte Carlo experiment uses the design dimensions

$$\begin{aligned} &n \in \{500, 1000\}, &&\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}, \\ &\kappa \in \{1, 0.50, 0.25, 0.10\}, &&R = 500. \end{aligned} \tag{50}$$

The Cartesian product contains 40 distinct (n, τ, κ) designs, each with 500 replications.

Oracle-GMM uses the known oracle control set; Full-GMM uses all controls without regularization; and DML-IVQR uses regularized high-dimensional nuisances and an orthogonalization-based score construction, with the implementation-specific residualization intercept described in Section 3.3. The design therefore has 120 design–estimator cells. Every comparison uses the truth $\alpha_0(\tau) = 1 + \Phi^{-1}(\tau)$ from Equation (43).

All estimators use the same sample within a replication. DML-IVQR uses the BC pivotal penalty and same-sample procedure in Section 3.3. Inference is conducted at 95%, and power uses the alternatives defined below.

4.3 Performance Measures and Numerical Inference

For a fixed $(n, \tau, \kappa, \text{estimator})$ cell, let $\widehat{\alpha}_r(\tau)$ denote the point estimate in replication r, and let $\alpha_0(\tau)$ be the true value from Equation (43). Replications are indexed by $r = 1, \dots, R$. Suppressing an estimator superscript, point-estimation performance is summarized by

$$\begin{aligned} \text{Bias}(\tau) &= \frac{1}{R} \sum_{r=1}^{R} \left[\alpha_0(\tau) - \widehat{\alpha}_r(\tau)\right], \\ \text{MAE}(\tau) &= \frac{1}{R} \sum_{r=1}^{R} \left|\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right|, \\ \text{RMSE}(\tau) &= \left\{\frac{1}{R} \sum_{r=1}^{R} \left[\widehat{\alpha}_r(\tau) - \alpha_0(\tau)\right]^2\right\}^{1/2}. \end{aligned} \tag{51}$$

Measures are computed separately by estimator and design cell. Positive Bias means estimates lie below the truth on average; MAE records absolute error, and RMSE gives greater weight to large errors.

Coverage is determined by evaluating the test statistic directly at the exact true parameter. With $c_{0.95}$ denoting the critical value defined below, define

$$C_r(\tau) = \mathbf{1}\left\{W_{N,r}(\alpha_0(\tau)) \le c_{0.95}\right\}, \qquad \widehat{\text{Coverage}}(\tau) = \frac{1}{R} \sum_{r=1}^{R} C_r(\tau). \tag{52}$$

Coverage evaluates $W_{N,r}$ directly at $\alpha_0(\tau)$ and is therefore independent of the numerical grid. Empirical size is

$$\widehat{\text{Size}}(\tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\left\{W_{N,r}(\alpha_0(\tau)) > c_{0.95}\right\} = 1 - \widehat{\text{Coverage}}(\tau). \tag{53}$$

The equality follows from the implemented rejection rule.

Power is evaluated at deliberately false treatment-effect values

$$a_{\Delta}(\tau) = \alpha_0(\tau) + \Delta, \qquad \Delta \in \{-1.00, -0.50, -0.25, 0.25, 0.50, 1.00\}. \tag{54}$$

For each offset, empirical power is

$$\widehat{\text{Power}}(\Delta, \tau) = \frac{1}{R}\sum_{r=1}^{R} \mathbf{1}\left\{W_{N,r}(a_{\Delta}(\tau)) > c_{0.95}\right\}. \tag{55}$$

The statistic is evaluated directly at each false value, independently of the confidence-region grid; $\Delta = 0$ is size, not power.

The accepted-set measure totals all accepted components inside the primary computational domain, including disconnected components. It is neither the span between extreme accepted values nor an unrestricted confidence-region length. Because $d_z = 2$, the 95% cutoff for the statistic in Section 3.4 is

$$c_{0.95} = \chi^2_{2,0.95} = 5.991464547\ldots. \tag{56}$$

A candidate is accepted when $W_N(a) \leq c_{0.95}$. This asymptotic cutoff is not an exact finite-sample guarantee. The primary grid is

$$\begin{aligned} A_0 &= [-1, 3], \qquad\qquad h = 0.1, \\ G_0 &= \{a_j = -1 + (j-1)h : j = 1, \ldots, 41\}, \\ &= \{-1, -0.9, \ldots, 2.9, 3\}. \end{aligned} \tag{57}$$

The same grid is used throughout. A_0 is a computational domain, not an asserted parameter space, so reaching its boundary establishes neither global boundedness nor unboundedness.

The initial numerical design considered reporting total confidence-region length after adaptive domain expansion. Numerical grid diagnostics showed that, under weak relevance, accepted sets could remain non-localized over increasingly wide numerical domains. The final design therefore reports a fixed-domain accepted-set measure and treats domain expansion as a separate numerical diagnostic rather than assigning an artificial finite confidence-region length.

For replication r, define the binary grid-acceptance indicator, accepted grid, and all-grid-points indicator by

$$\begin{aligned} I_r(a_j) &= \mathbf{1}\{W_{N,r}(a_j) \le c_{0.95}\}, \\ C_{r,G_0} &= \{a_j \in G_0 : I_r(a_j) = 1\}, \\ F_r &= \mathbf{1}\{I_r(a_j) = 1 \text{ for every } a_j \in G_0\}. \end{aligned} \tag{58}$$

The frozen implementation computes the grid-based accepted-set measure within A_0 as the trapezoidal integral of this binary indicator:

$$M_r(A_0) = \sum_{j=1}^{40} \frac{a_{j+1} - a_j}{2} \left[I_r(a_j) + I_r(a_{j+1})\right]. \tag{59}$$

The trapezoidal sum includes separated accepted components without filling rejected gaps and does not interpolate critical-value crossings. It ranges from zero to four. The Monte Carlo mean of F_r is the frequency with which all tested grid points are accepted, not acceptance of the real line.

Coverage and power remain direct evaluations outside this grid construction. A post-main diagnostic examines sensitivity to the finite endpoints using

$$A_0 = [-1, 3], \qquad A_1 = [-3, 5], \qquad A_2 = [-5, 7]. \tag{60}$$

It uses $n = 1000$, 100 saved primitive samples, all five quantiles and estimators, $\kappa \in \{1, 0.25, 0.10\}$, and spacing 0.1. Stage 1 evaluates A_1 and its nested A_0 profile; Stage 2 reloads the same samples and adds only the tails needed for A_2. Domain comparisons therefore change candidates, not samples. The outputs record accepted measure, grid share and count, boundary acceptance, all-grid acceptance, and incremental measure.

Boundary acceptance shows only a lack of localization within the examined domain; it does not establish global unboundedness, acceptance of the real line, or failed identification. The bounded diagnostic ends at A_2, while main summaries retain A_0 and direct coverage and power remain unchanged.

5 Monte Carlo Results

5.1 Instrument Relevance and Point Estimation

Table 1 confirms that κ creates four clearly ordered relevance settings. For $n = 500$, the median first-stage F-statistic falls from 101.17 at $\kappa = 1$ to 18.71 at $\kappa = 0.50$ and 1.22 at $\kappa = 0.10$;

Table 1. Instrument-strength calibration

n	κ	**Median** F	F**: p10–p90**	**Median partial** R^2	**Partial** R^2**: p10–p90**
500	1.00	101.17	85.93–126.68	0.294	0.261–0.342
500	0.50	18.71	11.99–27.95	0.071	0.047–0.103
500	0.25	4.74	1.74–8.65	0.019	0.007–0.034
500	0.10	1.22	0.24–3.65	0.005	0.001–0.015
1000	1.00	212.82	185.34–239.50	0.301	0.273–0.327
1000	0.50	38.65	30.13–53.60	0.073	0.058–0.098
1000	0.25	9.85	5.34–16.15	0.020	0.011–0.032
1000	0.10	2.04	0.38–5.42	0.004	0.001–0.011

Note: Descriptive Monte Carlo strength diagnostics from the preserved 100-replication calibration. The reported dispersion ranges are the 10th and 90th percentiles. These statistics are not formal weak-instrument classification thresholds.

the corresponding endpoints for $n = 1000$ are 212.82 and 2.04. The median partial R^2 is less sensitive to sample size: it is 0.294 and 0.301 at $\kappa = 1$, but only 0.005 and 0.004 at $\kappa = 0.10$. The excluded instruments therefore explain a progressively smaller share of treatment variation conditional on the controls. The stable partial-R^2 ordering across sample sizes shows that this decline comes from the designed signal, whereas the F-statistic also rises with the amount of sample information. These measures describe relative relevance within the experiment; they are not formal classifications of quantile-specific IVQR identification.

Figure 1 shows a corresponding deterioration in point estimation. RMSE rises as κ falls for every estimator, quantile, and sample size. When the instruments contain less information about treatment, the GMM criterion becomes less informative about which candidate value best matches the moment conditions, so weaker relevance leads to substantially larger point-estimation errors. This interpretation concerns accuracy and dispersion, not a mechanical shift in one direction. The estimates are the numerical profile minimizers defined in Sections 3.4 and 4.3 rather than unrestricted continuous optimizers.

At $\tau = 0.50$ and $n = 1000$, Table 2 provides a representative scale. From $\kappa = 1$ to $\kappa = 0.10$, RMSE rises from 0.072 to 0.945 for Oracle-GMM, from 0.100 to 1.018 for Full-GMM, and from 0.104 to 0.991 for DML-IVQR-BC. MAE moves in the same direction: the respective endpoint pairs are 0.047 and 0.735, 0.072 and 0.840, and 0.079 and 0.779. Signed bias is often much smaller than MAE and RMSE. At $\kappa = 0.25$, for example, biases of 0.045, 0.029, and −0.003 accompany MAEs between 0.281 and 0.372. Under the reporting convention, positive Bias means that estimates lie below the truth on average. The deterioration there-

Table 2. Point-estimation performance at the median quantile, $n = 1000$

κ	**Estimator**	**Bias**	**MAE**	**RMSE**
1.00	Oracle-GMM	0.007	0.047	0.072
1.00	Full-GMM	0.007	0.072	0.100
1.00	DML-IVQR-BC	0.070	0.079	0.104
0.50	Oracle-GMM	0.015	0.093	0.134
0.50	Full-GMM	0.014	0.151	0.199
0.50	DML-IVQR-BC	0.059	0.119	0.160
0.25	Oracle-GMM	0.045	0.281	0.431
0.25	Full-GMM	0.029	0.372	0.500
0.25	DML-IVQR-BC	-0.003	0.306	0.448
0.10	Oracle-GMM	-0.044	0.735	0.945
0.10	Full-GMM	-0.012	0.840	1.018
0.10	DML-IVQR-BC	-0.195	0.779	0.991

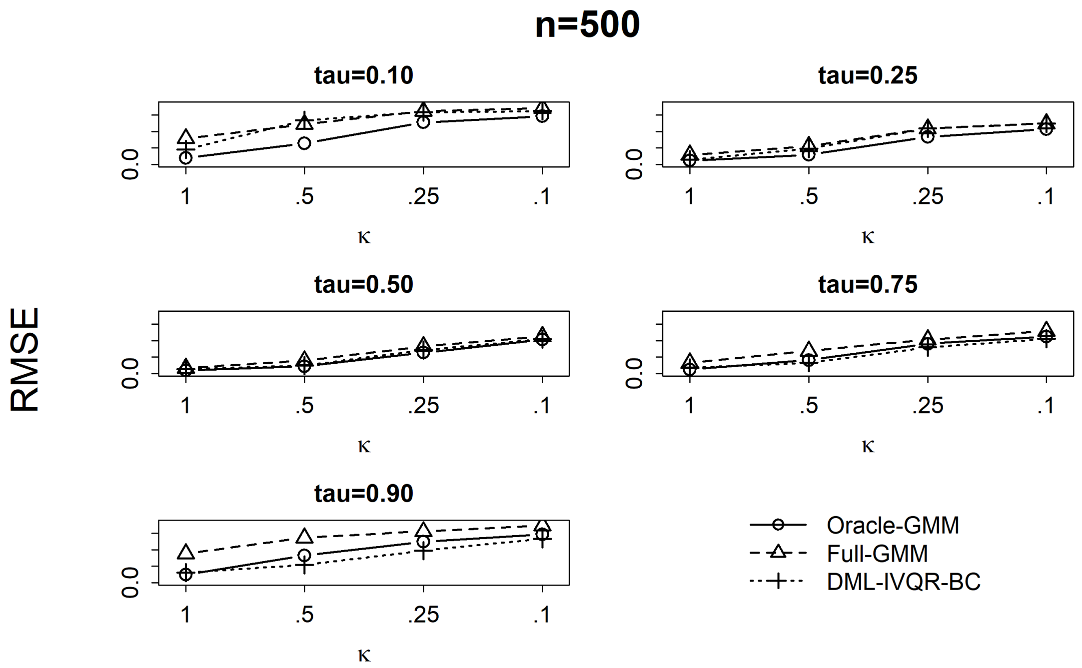
Note: Each cell uses 500 Monte Carlo replications. The frozen sign convention is Bias = $\alpha_0 - \widehat{\alpha}$.

fore reflects substantial replication-to-replication error rather than only a common directional shift.

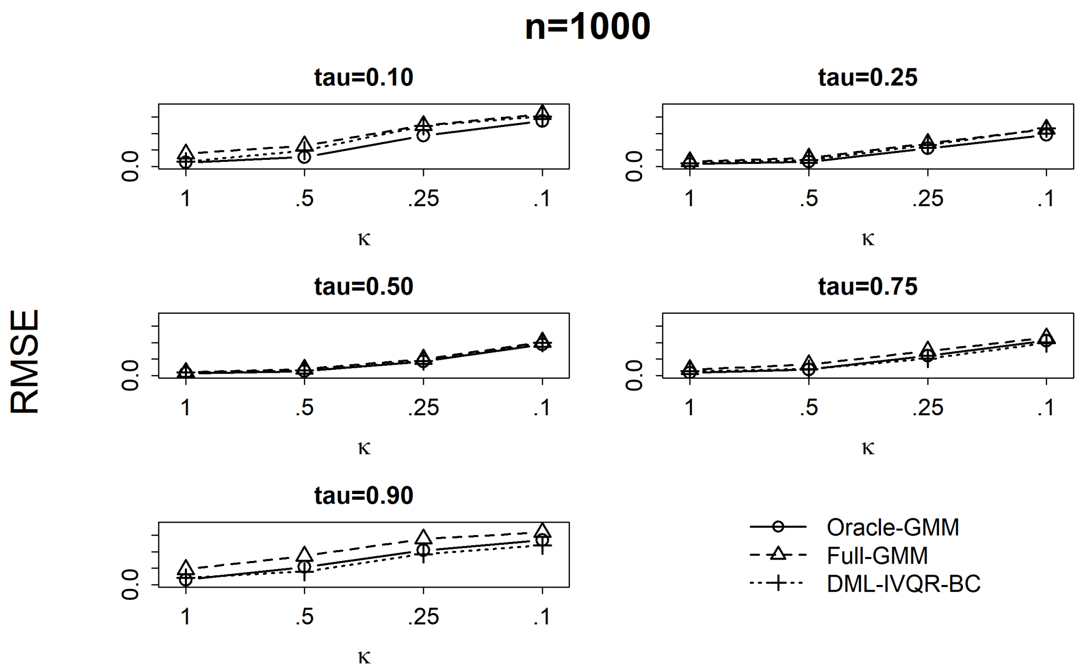
Oracle-GMM generally supplies the strongest benchmark because it knows the designated oracle control set, yet its error still rises sharply as relevance weakens. The shared deterioration cannot therefore be attributed only to estimating a high-dimensional nuisance model. Full-GMM often has higher RMSE than Oracle and, in many cells, higher RMSE than DML. DML is neither uniformly better than Full nor uniformly equal to Oracle, and the comparison does not isolate regularization because the procedures also differ in instrument residualization. For every matched cell, Figure 1 records lower RMSE at $n = 1000$ than at $n = 500$. More observations improve accuracy, but do not remove the pronounced loss at $\kappa = 0.10$.

5.2 Confidence-Set Informativeness and Power

Figures 2 and 3 show that the median accepted-set measure increases sharply as κ falls for all estimators. The measure is the total amount of accepted candidate values inside the four-unit primary domain A_0; it is not the unrestricted length of a theoretical confidence region. At $\kappa = 1$, its median is small relative to the domain maximum. At $\kappa = 0.10$, every estimator–quantile combination has a median of at least 3.8, so the tests retain nearly the whole primary domain.



(a) $n = 500$



(b) $n = 1000$

Figure 1. RMSE across instrument-strength designs

Note: Values of κ are plotted as equally spaced design categories in descending relevance order, from the strongest setting on the left to the weakest on the right.

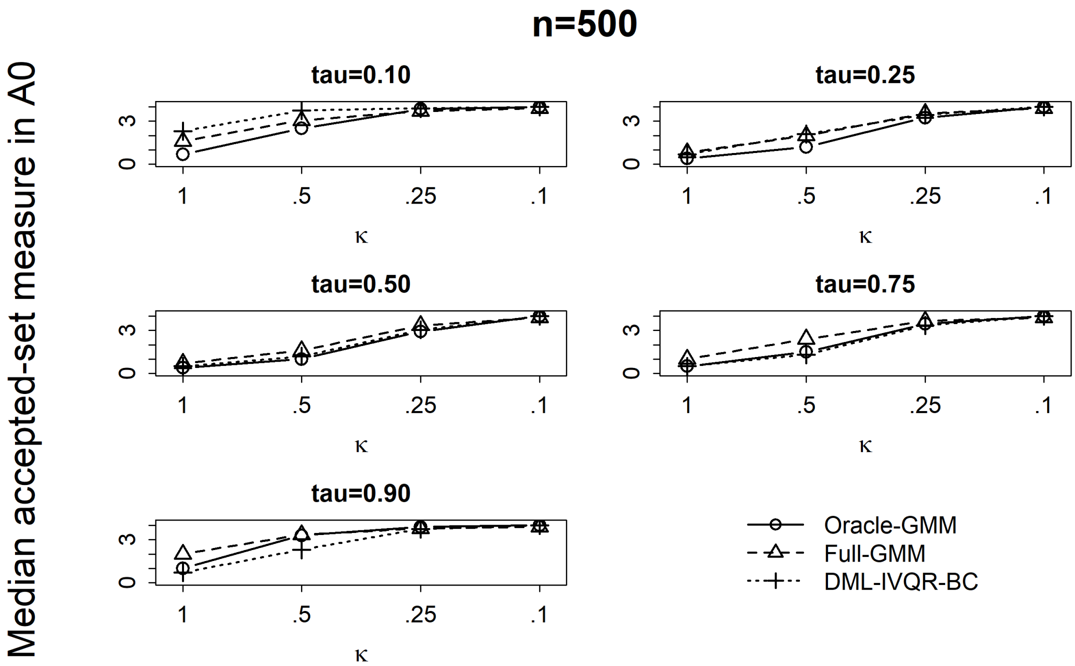


Figure 2. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, $n = 500$

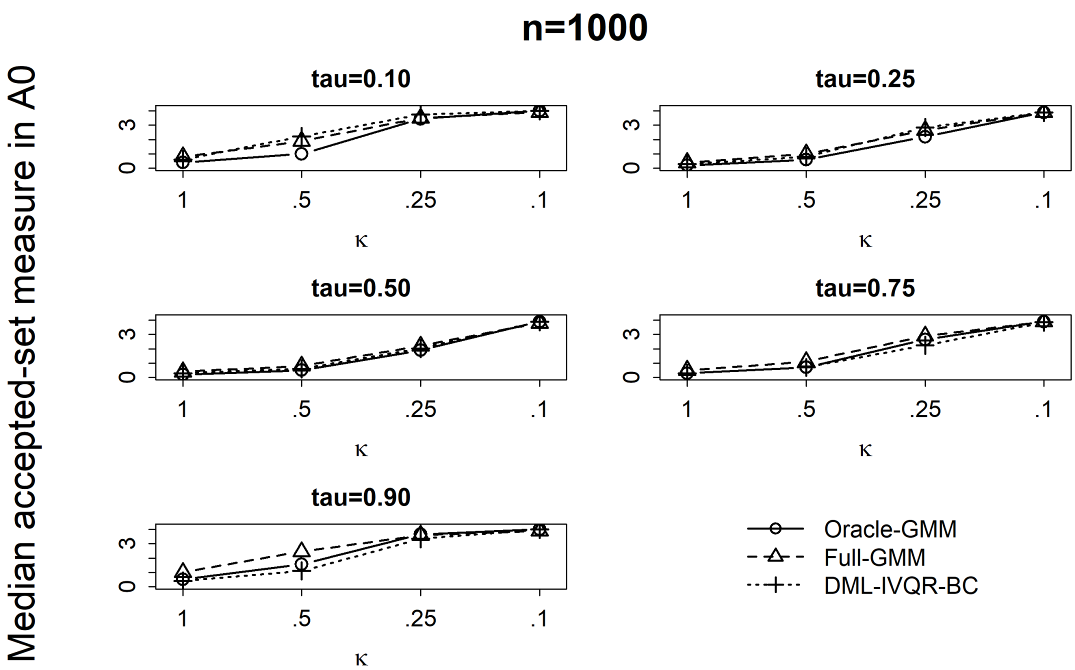


Figure 3. Grid-based accepted-set measure within the primary computational domain $A_0 = [-1, 3]$, $n = 1000$

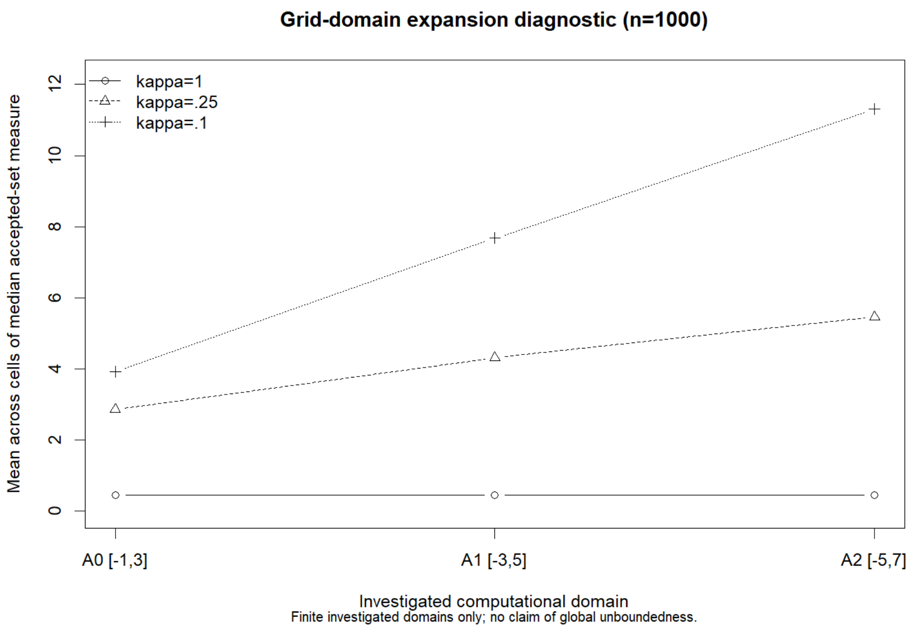


Figure 4. Sensitivity of the accepted set to expansion of the computational domain

Note: Each plotted value is the mean across 15 estimator–quantile cells of the cell-specific median accepted-set measure for the stated κ and domain. The investigated domains are $A_0 = [-1, 3]$, $A_1 = [-3, 5]$, and $A_2 = [-5, 7]$. The diagnostic records whether the accepted set stabilizes within these numerical domains; it does not establish global boundedness or unboundedness.

For $n = 1000$ and $\tau = 0.50$, the Oracle, Full, and DML medians are respectively 0.200, 0.400, and 0.300 at $\kappa = 1$; 1.900, 2.200, and 2.025 at $\kappa = 0.25$; and 3.900, 3.800, and 3.900 at $\kappa = 0.10$. Oracle often has the smallest set when relevance is informative, but no ordering is uniform. Increasing n from 500 to 1000 weakly reduces the median measure in every matched cell, without restoring localization at the weakest setting. The oracle's expansion shows again that knowledge of the relevant controls cannot compensate for little instrument information.

Figure 4 qualifies the finite-domain measure. Expanding A_0 to A_1 and A_2 adds little accepted measure at $\kappa = 1$, indicating numerical localization within the investigated domains. At $\kappa = 0.25$, expansion adds substantial measure in some profiles. In the median-quantile DML diagnostic, the median rises from 1.950 in A_0 to 2.500 in A_1 and 3.025 in A_2. At $\kappa = 0.10$, the corresponding sequence is 3.900, 7.500, and 10.325, and the aggregate measure does not stabilize within the investigated domains. This diagnostic shows continuing acceptance beyond A_0; it neither proves global unboundedness nor establishes failed identification. Appendix A records its scope and numerical handling.

Power supplies the direct rejection-probability counterpart. Figure 5 reports power at $\Delta = -0.50$ and $\Delta = +0.50$ for $n = 1000$. Rejection probabilities decline sharply as κ falls, apart from occasional small reversals when they are already low. At the median and $\Delta = -0.50$, Oracle power moves from 1.000 through 0.850 and 0.316 to 0.082; DML moves from 1.000 through 0.802 and 0.312 to 0.104. Full falls from 0.998 at the strongest setting to 0.076 at the weakest. For $\Delta = +0.50$, Oracle declines from 1.000 to 0.092, while weakest-setting power is 0.098 for Full and 0.094 for DML. Thus, by $\kappa = 0.10$ the tests provide little discrimination against these false values even at $n = 1000$.

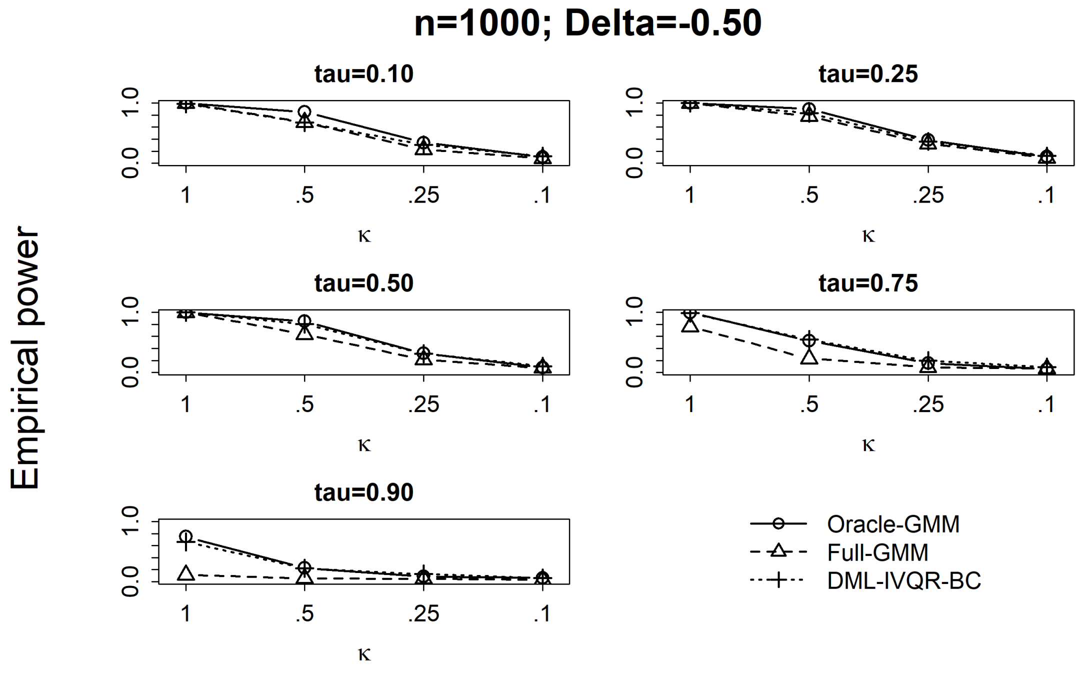
More distant alternatives are generally easier to reject, although power is not perfectly monotone in $|\Delta|$ in every finite-sample cell. Equal positive and negative offsets can also yield different rejection probabilities because candidate-specific IVQR moments and their finite-sample distributions need not be symmetric around the truth. Larger samples and Oracle-GMM often improve discrimination when usable signal remains, and DML frequently exceeds Full, but no estimator is uniformly most powerful. Expanding accepted sets and declining power describe the same loss of identifying information: more candidate values survive because their moments become harder to distinguish from those at the truth. This is reduced informativeness, not by itself a failure of test calibration.

5.3 Coverage and Finite-Sample Calibration

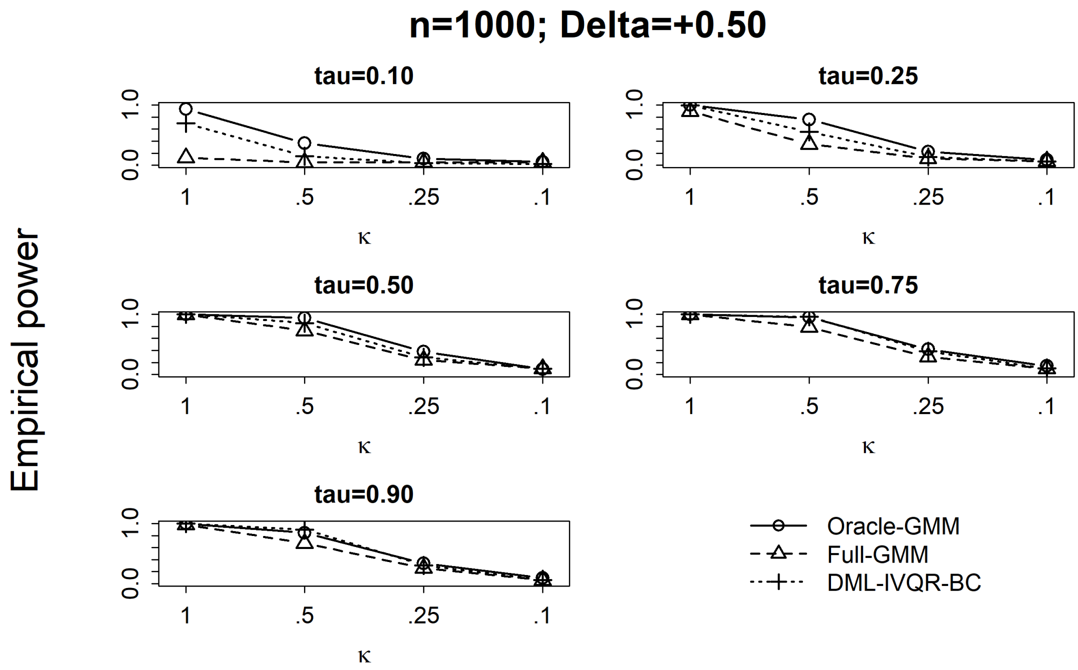
Coverage is evaluated directly at $\alpha_0(\tau)$, independently of the confidence-set grid; empirical size is its complement. The intended values are 0.95 and 0.05. Because the implemented statistic uses an asymptotic chi-square reference distribution, weak-identification robustness does not imply exact finite-sample calibration. Table 3 demonstrates substantial undercoverage even at $n = 1000$ and $\kappa = 1$. Full-GMM covers at 0.502 for $\tau = 0.10$ and 0.530 for $\tau = 0.90$; DML covers at 0.632 for $\tau = 0.75$ and 0.498 for $\tau = 0.90$. Oracle also undercovers, with 0.860 at both $\tau = 0.10$ and $\tau = 0.75$, so high-dimensional nuisance estimation is not the sole source of distortion. DML's 0.968 coverage at $\tau = 0.10$ simultaneously shows that the distortion is strongly quantile and estimator dependent.

The empirical 95th percentile of $W_N(\alpha_0)$ diagnoses the mismatch. The fixed cutoff is 5.991, whereas the Full percentiles are 21.090 at $\tau = 0.10$ and 20.072 at $\tau = 0.90$; the DML percentiles are 16.034 at $\tau = 0.75$ and 18.682 at $\tau = 0.90$. A cutoff far below the finite-sample null percentile rejects the truth too often. Conversely, DML's percentile of 5.256 at $\tau = 0.10$ is consistent with slight overcoverage. These percentiles are diagnostics, not replacement critical values.

Across all 120 design–estimator cells, Figures 6 and 7 show coverage ranging from 0.442 to 0.976. The forensic audit in Appendix B finds that 54 Monte Carlo 95% intervals exclude



(a) $\Delta = -0.50$



(b) $\Delta = +0.50$

Figure 5. Rejection probability under moderate false alternatives, $n = 1000$

Table 3. Finite-sample coverage under the strongest instrument design

τ	Estimator	Coverage	Size	MCSE	$q_{.95}(W_{\text{true}})$
0.10	Oracle-GMM	0.860	0.140	0.016	9.723
0.10	Full-GMM	0.502	0.498	0.022	21.090
0.10	DML-IVQR-BC	0.968	0.032	0.008	5.256
0.25	Oracle-GMM	0.874	0.126	0.015	8.300
0.25	Full-GMM	0.790	0.210	0.018	10.940
0.25	DML-IVQR-BC	0.894	0.106	0.014	7.811
0.50	Oracle-GMM	0.872	0.128	0.015	8.511
0.50	Full-GMM	0.896	0.104	0.014	7.524
0.50	DML-IVQR-BC	0.776	0.224	0.019	11.157
0.75	Oracle-GMM	0.860	0.140	0.016	8.763
0.75	Full-GMM	0.790	0.210	0.018	12.489
0.75	DML-IVQR-BC	0.632	0.368	0.022	16.034
0.90	Oracle-GMM	0.904	0.096	0.013	7.395
0.90	Full-GMM	0.530	0.470	0.022	20.072
0.90	DML-IVQR-BC	0.498	0.502	0.022	18.682

Note: Results use $n = 1000$, $\kappa = 1$, and 500 Monte Carlo replications. The unchanged 95% reference cutoff is $q_{\chi^2_2}(.95) = 5.991$.

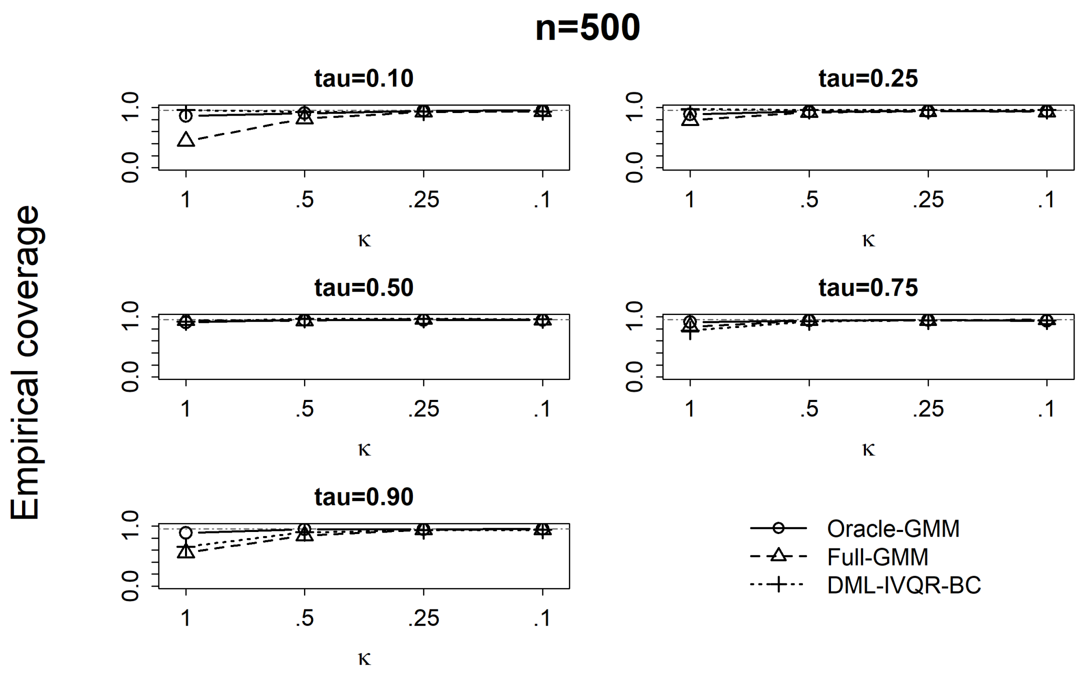


Figure 6. Empirical coverage across instrument-strength designs, $n = 500$

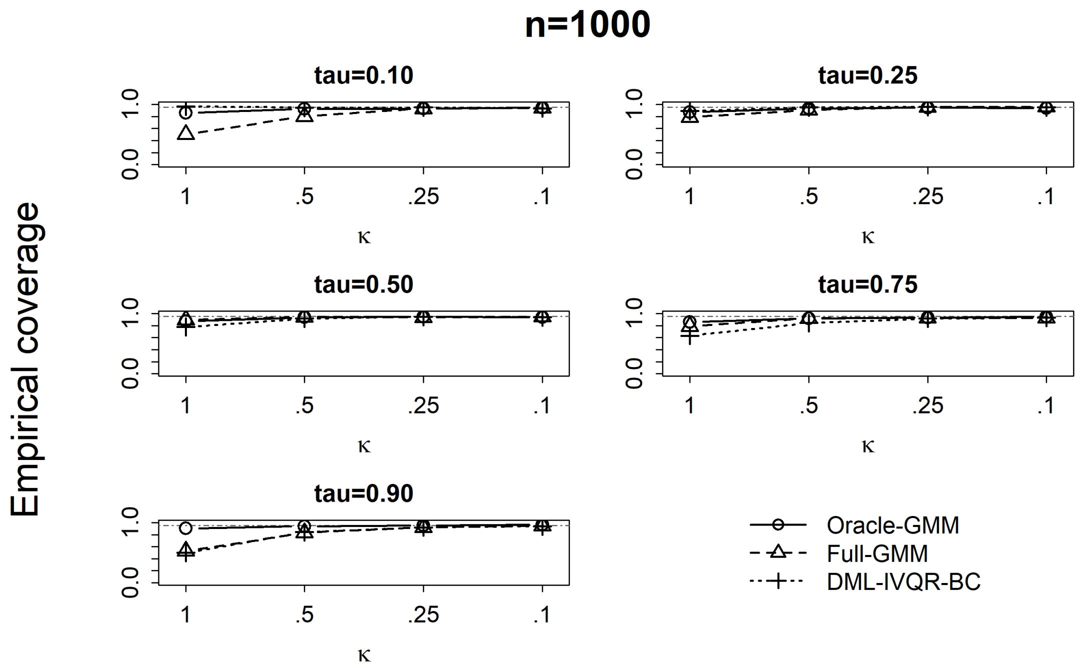


Figure 7. Empirical coverage across instrument-strength designs, $n = 1000$

nominal 0.95, confirms direct evaluation at the truth, and rules out missing or nonfinite statistics as the explanation. Coverage often rises as κ approaches 0.10, but neither a larger sample nor a particular estimator moves it uniformly toward the target.

Higher coverage under very low relevance is not automatically better inference. The truth can survive frequently because the test rejects little, while accepted sets cover most of the numerical domain and power against false values is low. Conversely, undercoverage means that the truth is rejected too often. Coverage, informativeness, and power must therefore remain separate. The observed distortions concern the finite-sample quality of the asymptotic approximation at the simulated sample sizes, not the validity of its limiting result.

5.4 Quantile Heterogeneity and Estimator Comparison

Weakening relevance affects every quantile adversely, but not by the same amount. RMSE is often larger in the tails than near the center, especially for Full-GMM: at $n = 1000$ and $\kappa = 1$, its RMSE is 0.100 at the median and 0.465 at $\tau = 0.90$. This ordering is not universal. Accepted sets are generally smaller around the middle under strong relevance, but at $\kappa = 0.10$ cross-quantile differences become modest relative to the common near-domain-wide acceptance.

Power reveals pronounced directional asymmetry. For DML at $n = 1000$ and $\kappa = 0.50$, power against $\Delta = -0.50$ is 0.678 at $\tau = 0.10$ and 0.224 at $\tau = 0.90$; against $\Delta = +0.50$, the respective values are 0.148 and 0.900. Neither tail is therefore uniformly more informative. Coverage has a different pattern: in the strong-relevance table, DML coverage moves from 0.968 at $\tau = 0.10$ to 0.776 at the median and 0.498 at $\tau = 0.90$. Such heterogeneity can reflect the quantile-varying structural effect, score, local outcome density, effective sample information, and nuisance difficulty, but the experiment does not isolate these channels.

Across estimators, Oracle usually gives the strongest benchmark but still deteriorates sharply with relevance. DML often records lower error, smaller accepted-set measures, and higher power than Full, consistent with the importance of how high-dimensional nuisances are estimated. It does not dominate Full in every cell, reproduce Oracle performance uniformly, or achieve uniformly accurate coverage. Because the procedures also differ in residualization details, these comparisons do not identify the causal effect of regularization alone.

The combined evidence answers the research question: weaker relevance raises point-estimation error, expands accepted sets, and reduces power for every estimator, while the magnitude and calibration consequences vary across quantiles. Regularized nuisance estimation can mitigate part of the high-dimensional cost, but cannot prevent the loss of structural information when the instruments contain little information about treatment. The oracle comparison establishes the common relevance problem; the comparison with Full shows that nuisance estimation still matters in finite samples.

6 Discussion

6.1 Weak Relevance, Identification, and Inferential Informativeness

The common deterioration across Oracle-GMM, Full-GMM, and DML-IVQR-BC is consistent with reduced identifying information. In IVQR, identification depends on whether instrument-induced treatment variation changes the structural quantile moments enough to separate $\alpha_0(\tau)$ from candidate values. When relevance is high, moving a away from the truth tends to move the population moment away from zero, and $W_N(a)$ can reveal that disagreement. As relevance declines, true and false candidates generate more similar moment behavior, though an individual sample profile need not be literally flat. The same mechanism explains three findings: point-estimation error rises, more candidates remain in the inverted accepted set, and false candidates are rejected less often. The design parameter κ creates this controlled change in relevance; it is not itself a formal measure of IVQR identification.

Oracle-GMM shows that high-dimensional nuisance estimation is not sufficient to explain the pattern. It knows the designated oracle control set but still experiences increasing error, expanding accepted sets, and declining power. Estimator-specific nuisance and residualization details prevent a perfect decomposition, yet the component shared by all three procedures is weaker information from the excluded instruments. Increasing the sample from 500 to 1000 generally improves accuracy and discrimination, but does not eliminate the differences across κ. Quantile-specific rates and asymmetries also remain, consistent with the density-weighted nature of IVQR information, although the simulation does not construct a formal quantile-specific identification statistic.

The domain diagnostic reinforces this interpretation while limiting the claim appropriately. Expansion adds little accepted measure at $\kappa = 1$, substantial measure in some profiles at $\kappa = 0.25$, and continuing measure through the largest investigated domain at $\kappa = 0.10$. The weakest designs therefore provide little numerical localization within the investigated ranges. They are not thereby proven globally unbounded or formally unidentified.

Robustness and informativeness answer different questions. The candidate-specific procedure of Chernozhukov and Hansen (2008) avoids relying on an approximately normal, strongly identified point estimator. Its limiting null result can remain valid under weak, partial, or failed identification, but it does not promise narrow confidence regions or high power. A wide or disconnected region can correctly report that the maintained model and data exclude few values. Finite-sample calibration is separate again. The preserved procedure compares $W_N(a)$ with an asymptotic χ^2_2 cutoff, and Chapter 5 shows that its finite-sample null distribution can differ substantially from this reference. Direct coverage and the audit rule out grid artifacts, while Oracle undercoverage shows that the issue need not originate in high-dimensional selection. The evidence concerns approximation quality at the simulated sample sizes, not the validity of the limiting theorem. A procedure may consequently be asymptotically robust but uninformative, poorly calibrated in finite samples, or both.

6.2 What DML Can and Cannot Address

DML-IVQR targets estimation of high-dimensional nuisance parameters surrounding the low-dimensional structural coefficient. Quantile Lasso regularizes the candidate-specific outcome nuisance, and regularized weighted regressions estimate instrument residualization. Neyman orthogonality reduces the first-order effect of sufficiently small nuisance-estimation errors on the population moment (Chernozhukov et al. 2018; Chen et al. 2021). This local property neither eliminates nuisance error nor guarantees accurate finite-sample chi-square calibration. More fundamentally, it does not change the variation supplied by the instruments.

The comparison with Full-GMM indicates why nuisance treatment can matter. Full uses all controls without ℓ_1 regularization, while DML often has lower RMSE, smaller accepted sets, and higher power. The ordering varies across quantiles, relevance settings, and metrics, so DML is not uniformly preferable. Nor can the differences be attributed entirely to regularization: DML and Full also differ in their nuisance and residualization procedures, including intercept treatment. The comparison supports conclusions about the implemented estimators, not a causal decomposition of their components.

Oracle-GMM helps delimit this explanation. By avoiding estimation of the full high-dimensional nuisance model, it removes the central problem that DML is designed to manage, yet it deteriorates with κ alongside the other procedures. Orthogonality changes the influence of nuisance-estimation error; it cannot turn weakly informative structural moments into strongly informative ones. The common low-relevance deterioration is therefore not evidence against regularization. It demonstrates that nuisance robustness and instrument relevance are separate requirements.

The DML label here refers to the preserved DML-IVQR-BC implementation. It uses the pivotal penalty, fitted-normal density proxy, and same-sample score evaluation described in Section 3.3; it does not use cross-fitting. These choices define the comparison without implying that their alternatives would improve performance. The results consequently support a bounded conclusion: handling a large nuisance model can improve finite-sample behavior relative to the unregularized full-control procedure, but no nuisance method can restore structural information removed from the instruments.

6.3 Limitations and Implications

The experiment is based on one high-dimensional IVQR DGP derived from Chen et al. (2021). Holding its treatment distribution, endogeneity channel, outcome equation, structural quantile effect, controls, and instruments fixed across κ strengthens the internal interpretation, but limits generalization to other disturbance distributions, degrees of endogeneity, instrument dimensions, sparsity patterns, or structural functions. The two sample sizes do not describe very small samples, larger-sample convergence, or rates. Five quantiles and six power offsets reveal heterogeneity but do not map the continuous quantile or power functions. Monte Carlo summaries based on 500 replications retain simulation uncertainty, which the coverage audit treats explicitly.

The evidence also concerns one same-sample DML implementation rather than all possible nuisance learners or cross-fitting schemes. The three estimators differ in more than regularization, so their comparison does not isolate any one component. Inference uses the preserved asymptotic chi-square cutoff; bootstrap corrections and the exact conditional procedure of Chernozhukov et al. (2009) are not evaluated. The accepted-set measure is computed on a finite grid, and even the expanded domains cannot establish global boundedness or unboundedness. Finally, the thesis contains no empirical application, so it does not measure the frequency or importance of these patterns in a particular economic setting.

These limitations also shape empirical practice. A conventional first-stage diagnostic cannot by itself establish quantile-specific identification, and nominal coverage says little about how many economically distinct effects remain plausible. Reporting the criterion or accepted region over a transparent candidate domain, together with power or sensitivity calculations where feasible, makes the information content visible. When calibration is doubtful, alternative critical-value procedures should be evaluated rather than assuming that orthogonal nuisance estimation resolves the approximation problem. Such reporting does not cure weak relevance, but it separates evidence about validity, numerical scope, and discrimination.

These boundaries define the scope of the conclusions. Within the preserved DGP and implementation, weaker relevance produces larger point-estimation errors, less informative accepted sets, and lower power, with material quantile heterogeneity and imperfect finite-sample calibration in some cells. The practical implication is to report instrument information, point accuracy, coverage, and inferential informativeness separately rather than treating robust test inversion or DML as evidence of precision. Future work could compare alternative DGPs and instrument structures, larger samples, cross-fitted or other nuisance estimators, finite-sample critical values, and empirical applications.

7 Conclusion

This thesis asks how systematically weakening excluded-instrument relevance affects the finite-sample performance and informativeness of weak-identification-robust inference in DML-IVQR, and how these effects vary across quantiles. The Monte Carlo experiment changes relevance through κ while preserving the treatment distribution, endogeneity channel, structural outcome equation, quantile treatment effect, controls, and observed instruments of the baseline design in Chen et al. (2021). Oracle-GMM, Full-GMM, and DML-IVQR-BC are evaluated on the same samples.

As excluded-instrument relevance declines, point-estimation error rises substantially for all three estimators. Oracle-GMM knows the designated oracle control set and avoids estimating the complete high-dimensional nuisance model, yet its performance deteriorates alongside the full-control procedures. The common pattern therefore cannot be explained only by regularization, model selection, or estimation of many nuisance coefficients. Reduced information supplied by the instruments is central.

Inference becomes less informative for the same reason. The accepted-set measure grows toward the maximum of the primary computational domain, domain expansion reveals continuing acceptance beyond that domain in weak-relevance profiles, and power against false candidates falls sharply. These findings do not establish that theoretical confidence regions are globally unbounded or that identification formally fails. They show that the candidate-specific moments provide little numerical discrimination within the investigated ranges. Larger samples generally improve accuracy and rejection power, but do not eliminate this deterioration at the two sample sizes studied. Its magnitude also varies across quantiles, and neither tail is uniformly most difficult.

Finite-sample calibration is a distinct qualification. Coverage ranges from 0.442 to 0.976, and 54 of 120 Monte Carlo confidence intervals for coverage exclude the nominal value. Substantial undercoverage occurs in some strong-relevance cells and for Oracle-GMM, so it cannot be attributed solely to high-dimensional nuisance estimation. These results concern the finite-sample accuracy of the asymptotic chi-square approximation, not the validity of the limiting weak-identification-robust theory. Conversely, high coverage under very low relevance can coexist with accepted sets spanning most of the investigated domain and low power. Coverage alone therefore does not establish informative inference.

The estimator comparison shows both the value and the boundary of DML. DML-IVQR-BC often improves on unregularized Full-GMM, indicating that the treatment of high-dimensional nuisance parameters matters. It neither dominates in every cell nor reproduces Oracle performance uniformly, and implementation differences prevent attributing every gap to regularization alone. Most importantly, orthogonality reduces sensitivity to nuisance-estimation error but cannot create instrument relevance or identifying information.

The conclusions are limited to one preserved DGP, two sample sizes, one same-sample DML implementation, finite numerical domains, and asymptotic chi-square inference; no empirical application is undertaken. Future work should examine alternative instrument structures, larger samples, cross-fitted nuisance estimation, and finite-sample critical values. The central lesson is that nuisance robustness and weak-identification robustness are valuable but distinct from informativeness: when instruments contain little information about treatment, a robust high-dimensional IVQR procedure may still estimate imprecisely and exclude few false structural effects.

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Appendix A Numerical and Domain Diagnostics

The post-main diagnostic expands the candidate domain from $A_0 = [-1, 3]$ to $A_1 = [-3, 5]$ and then to $A_2 = [-5, 7]$. At $\kappa = 1$, the accepted-set measure changes little across these domains, supporting numerical localization within the investigated range. At $\kappa = 0.25$, expansion adds material accepted measure in some profiles. At $\kappa = 0.10$, the aggregate measure does not stabilize within A_2. These findings concern only the finite domains investigated and do not establish global unboundedness. Figure 4 provides the compact graphical summary, while the complete machine-readable diagnostic outputs remain preserved with the artifacts documented in Appendix C.

The diagnostic recorded one Stage 1 numerical failure: replication 35 at $\tau = 0.90$, $\kappa = 0.10$, for DML-IVQR-BC at candidate $a = -2.1$, with a singular-design error. Stage 2 recorded no new evaluation errors. The affected profile was treated as incomplete, and its aggregate accepted-set measures were not calculated from partial information.

Appendix B Coverage Forensic Audit

The coverage audit evaluates each estimator directly at the exact true value $\alpha_0(\tau)$ rather than inferring coverage from the finite confidence-region grid. It contains 60,000 direct evaluations: 500 replications for each of 120 sample-size, quantile, instrument-strength, and estimator cells. Consequently, the reported coverage results do not depend on whether the true value lies on the numerical candidate grid.

Empirical coverage ranges from 0.442 to 0.976. Wilson 95% Monte Carlo intervals exclude nominal 0.95 coverage in 54 of the 120 cells. The audit finds no missing or nonfinite true-parameter evaluations, so numerical failure or missing-value handling does not explain the documented undercoverage. The complete machine-readable audit outputs and their manifest remain preserved with the artifacts documented in Appendix C.

Appendix C Implementation and Reproducibility

This appendix records the repository materials needed to understand and reproduce the computational workflow. All paths are relative to the repository root. The archived outputs and manifests permit verification of the reported evidence, but the material is not described as bit-for-bit reproducible across arbitrary systems.

C.1 Preserved Source and Extension Architecture

The authors' source code remains unmodified in `Empirical_work/`, `example/`, and `simulation/`. Thesis-specific code is separated into `thesis_extension/src/`, with the frozen design in `thesis_extension/config/extension_config.R`. The wrapper files `author_oracle_wrapper.R`, `author_` `full_wrapper.R`, and `author_bc_dml_wrapper.R` reproduce the estimator logic used by the extension without modifying the preserved source directories. Main outputs are stored by sample size under `thesis_extension/final/output/n500/` and `thesis_extension/final/output/n1000/`. Diagnostic and reporting code occupy separate namespaces under `thesis_extension/diagnostics/` and `thesis_extension/reporting/`.

C.2 Computational Environment

The canonical runtime is R 3.4.3 dated 30 November 2017 on 64-bit Windows, and the simulations use five workers. The complete captured session is in `thesis_extension/environment/sessionInfo_R343.txt`, while `thesis_extension/environment/library_manifest_R343.csv` records the installed library inventory. The attached packages in the captured session are `doSNOW` 1.0.16, `snow` 0.4-2, `iterators` 1.0.8, `foreach` 1.4.3, `mvtnorm` 1.0-6, `hqreg` 1.4, `hdm` 0.2.0, `quantreg` 5.34, and `SparseM` 1.77. The session record also lists the loaded namespace versions, including `Matrix` 1.2-12, `MatrixModels` 0.4-1, and `Formula` 1.2-2.

C.3 Frozen Simulation Configuration

The main design uses $n \in \{500, 1000\}$, $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, $\kappa \in \{1, 0.50, 0.25, 0.10\}$, and 500 Monte Carlo replications per design. The estimators are Oracle-GMM, Full-GMM, and DML-IVQR-BC. Power is evaluated at offsets $\Delta \in \{-1, -0.5, -0.25, 0.25, 0.5, 1\}$. The primary candidate grid contains 41 values from -1 to 3 in increments of 0.1. Inference uses a 95% confidence level and the 0.95 quantile of a chi-square distribution with two degrees of freedom. The DML quantile-Lasso penalty follows the BC pivotal route with 1,000 simulated draws, scale factor 2, and penalty-tail probability 0.1. These settings are recorded in `thesis_extension/config/extension_config.R` and summarized in `thesis_extension/reporting/EMPIRICAL_RESULTS_FREEZE.md`.

C.4 Randomness and Seeds

The main simulation seed is 675, set in `thesis_extension/final/run_final.R` from the frozen extension configuration. The grid-domain diagnostic separately sets seed 675 in `thesis_extension/diagnostics/grid_sensitivity/grid_sensitivity_config.R`; its Stage 2 reuses the saved Stage 1 primitives rather than generating replacement samples. The 100-replication instrument-strength calibration uses pilot seed 20260813, recorded in `thesis_extension/calibration/instrument_strength_100rep/run_strength_calibration.R`. No distinct seed is asserted for a task unless it is recorded in these repository files.

C.5 Reproduction Workflow

This workflow documents how the archived analysis was produced and how it can be regenerated from the repository; it is not a one-command or bit-for-bit reproduction claim. Begin from the repository root with the canonical R 3.4.3 `Rscript` executable and the preserved package library described above. The reproduction workflow should be run from a clean or deliberately prepared output location. The simulation scripts do not provide a general safeguard against overwriting existing output files, so canonical archived results should be preserved before rerunning the workflow. The frozen manifests record the archived file hashes and can be used to compare newly generated outputs, but they do not constitute a general locking mechanism.

After the environment and output locations have been prepared, the empirical inputs are generated in the following dependency order:

1. Run the main simulation through `thesis_extension/final/run_final.R` with `--sample-size=500` and `--execute-final`, followed separately by `--sample-size=1000` and `--execute-final`.
2. Generate the reported instrument-strength calibration with `thesis_extension/calibration/instrument_strength_100rep/run_strength_calibration.R` and `--execute-calibration`.
3. Run the grid-domain diagnostic through `thesis_extension/diagnostics/grid_sensitivity/run_grid_sensitivity.R`, first with `--execute-stage1` and then, after checking the prespecified trigger, with `--execute-stage2`.
4. Run the coverage forensic audit through `thesis_extension/diagnostics/coverage_audit/run_coverage_audit.R` with `--execute-coverage-audit`; this audit reads the completed main-run outputs.
5. Only after all preceding empirical inputs exist, run `thesis_extension/reporting/build_thesis_results.R` with `--build-reporting`. The reporting builder reads the main-run, calibration, grid-diagnostic, coverage-audit, validation, and environment records; it does not rerun estimation or simulation.
6. Review the generated reporting consistency checks and the empirical-input and reporting-output manifests together with the diagnostic provenance records listed below.

For the frozen reporting build, the repository must be checked out at commit `ac9a9ccad027cde05488abb633e66cc02f87cc46`; the builder verifies this exact `HEAD` value but does not itself certify that the worktree is clean. All required empirical inputs must already be present. The directories `thesis_extension/reporting/output/` and `thesis_extension/reporting/figures/` must be empty or deliberately prepared as empty, and `EMPIRICAL_RESULTS_MANIFEST.csv` and `REPORTING_OUTPUT_MANIFEST.csv` must be absent from the reporting directory, because the builder refuses to replace nonempty reporting directories or existing reporting manifests.

C.6 Failure Handling

The canonical main-run summaries record zero numerical failures, as confirmed by `thesis_extension/reporting/output/reporting_consistency_checks.csv`. The grid-domain diagnostic records one Stage 1 failure: replication 35, $\tau = 0.90$, $\kappa = 0.10$, DML-IVQR-BC, and candidate $a = -2.1$, with the message "`Error info = 95 in stepy2: singular design.`" Stage 2 records no new evaluation failure. Candidate evaluation is wrapped in error handling: an error or nonfinite statistic is stored with its replication, design, estimator, candidate, stage, and message; the affected statistic is set to missing. A profile containing a failed evaluation is flagged as incomplete, and its aggregate accepted-set measures are reported as missing rather than being computed from a partial profile.

C.7 Theory and Code Correspondence

The preserved DML implementation estimates nuisance quantities and evaluates scores on the same sample; it does not use sample splitting or cross-fitting. It uses the BC pivotal penalty and the fitted-normal density proxy documented in Section 3.3. The literal density call is `dnorm(e, mean(e), var(e))`, so the residual sample variance is supplied to the argument that R interprets as a standard deviation. Instrument residualization regresses $\sqrt{\widehat{f}_i} Z_{ji}$ on transformed controls with an ordinary unpenalized intercept. The score subtracts that retained intercept and the fitted slopes from the untransformed instrument, without dividing the intercept by $\sqrt{\widehat{f}_i}$. As explained in Section 3.3, this intercept is implementation specific and is not the literal sample counterpart of the intercept in the augmented population projection.

C.8 Provenance and Verification Records

The empirical-results freeze is identified by commit `ac9a9ccad027cde05488abb633e66cc02f87cc46`, dated 2 September 2026; the pre-writing integration checkpoint is commit `911e0861114426b5aad314a6f2b7f0c1f92b3fcd`. The primary verification records are `PROJECT_STATUS.md`; `thesis_extension/validation/R343_final_audit/FINAL_PRE_RUN_AUDIT.md`; `thesis_extension/reporting/EMPIRICAL_RESULTS_FREEZE.md`; the input and output hash inventories `EMPIRICAL_RESULTS_MANIFEST.csv` and `REPORTING_OUTPUT_MANIFEST.csv` in the reporting directory; `thesis_extension/reporting/output/reporting_consistency_checks.csv`; and `thesis/figures/THESIS_FIGURE_DATA_CHECK.csv`. The grid diagnostic is documented by `GRID_RESULTS_PROVENANCE.md` and `GRID_RESULTS_MANIFEST.csv` in its directory, while the coverage check is documented by `COVERAGE_AUDIT_REPORT.md` and `COVERAGE_AUDIT_MANIFEST.csv`. Together these files record configuration, environment, source roles, output checks, and file hashes used to connect the archived computations to the reported thesis results.

Declaration

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

September 17, 2026

Olimjon Umurzokov